

07_Data Dictionary

Data Dictionary - GeoFare Database

1. PASSENGER Table

Column Name	Data Type	Size	Allow Null	Default	Description
passenger_id	VARCHAR	50	NO	-	Unique identifier for each passenger
name	VARCHAR	100	NO	-	Passenger's full name
contact_number	VARCHAR	20	NO	-	Mobile phone number (Philippine format)
email	VARCHAR	100	YES	NULL	Email address (optional)
password	VARCHAR	255	NO	-	Hashed password using bcrypt
created_at	DATETIME	-	NO	CURRENT_TIMESTAMP	Account creation date and time
updated_at	DATETIME	-	YES	NULL	Last update timestamp

2. TRIP Table

Column Name	Data Type	Size	Allow Null	Default	Description
trip_id	VARCHAR	50	NO	-	Unique identifier for each trip

Column Name	Data Type	Size	Allow Null	Default	Description
passenger_id	VARCHAR	50	YES	NULL	References passenger who took the trip
pickup_lat	DECIMAL	10,8	NO	-	Pickup point GPS latitude
pickup_lng	DECIMAL	11,8	NO	-	Pickup point GPS longitude
dropoff_lat	DECIMAL	10,8	NO	-	Drop-off point GPS latitude
dropoff_lng	DECIMAL	11,8	NO	-	Drop-off point GPS longitude
distance	DECIMAL	10,2	NO	-	Distance in kilometers
computed_fare	DECIMAL	10,2	NO	-	Estimated fare amount in PHP
trip_date_time	DATETIME	-	NO	CURRENT_TIMESTAMP	Trip start date and time
status	VARCHAR	20	YES	'Completed'	Current trip status
route_path	TEXT	-	YES	NULL	LINESTRING geometry of route
total_points	INT	-	YES	0	Number of GPS points recorded
total_distance	DECIMAL	10,2	YES	0	Total distance traveled (GPS log)

Status Values:

- 'In Progress' - Trip is ongoing
- 'Completed' - Trip is finished

- 'Reported' - Passenger filed a report
 - 'Cancelled' - Trip was cancelled
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3. GPS_LOG Table

Column Name	Data Type	Size	Allow Null	Default	Description
log_id	INT	-	NO	AUTO_INCREMENT	Unique identifier for each GPS log
trip_id	VARCHAR	50	NO	-	References the trip
latitude	DECIMAL	10,8	NO	-	GPS latitude coordinate
longitude	DECIMAL	11,8	NO	-	GPS longitude coordinate
timestamp	DATETIME	-	NO	CURRENT_TIMESTAMP	Time GPS point was captured

4. REPORT Table

Column Name	Data Type	Size	Allow Null	Default	Description
report_id	VARCHAR	50	NO	-	Unique identifier for each report
trip_id	VARCHAR	50	YES	NULL	References the trip being reported
reference_number	VARCHAR	20	NO	-	Public reference number (GEO-YYYYMMDD-XXXX)

Column Name	Data Type	Size	Allow Null	Default	Description
plate_number	VARCHAR	20	NO	-	Plate number (OCR or manual entry)
complaint_type	VARCHAR	50	NO	-	Type of complaint filed
description	TEXT	-	NO	-	Detailed description of the incident
status	VARCHAR	30	YES	'Pending'	Current report status
date_submitted	DATETIME	-	NO	CURRENT_TIMESTAMP	Report submission date and time
updated_by	VARCHAR	50	YES	NULL	LGU staff who last updated the report
updated_date	DATETIME	-	YES	NULL	Date and time of last update
resolution_notes	TEXT	-	YES	NULL	Notes when report is resolved
dismissal_reason	TEXT	-	YES	NULL	Reason if report is dismissed

Complaint Types:

- 'Overcharging'
- 'Safety Concern'
- 'Other'

Status Values:

- 'Pending' - Awaiting LGU review
 - 'Under Investigation' - Being investigated
 - 'Resolved' - Issue has been resolved
 - 'Dismissed' - Report was dismissed
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5. LGU_STAFF Table

Column Name	Data Type	Size	Allow Null	Default	Description
staff_id	VARCHAR	50	NO	-	Unique identifier for each LGU staff
name	VARCHAR	100	NO	-	Staff member's full name
username	VARCHAR	50	NO	-	Login username (unique)
password	VARCHAR	255	NO	-	Hashed password using bcrypt
role	VARCHAR	30	YES	'Staff'	User access level
status	VARCHAR	20	YES	'Active'	Account status
created_at	DATETIME	-	NO	CURRENT_TIMESTAMP	Account creation date
updated_at	DATETIME	-	YES	NULL	Last update timestamp

Roles:

- 'Admin' - Full system access including fare rates
- 'Staff' - Can view and update reports

Status Values:

- 'Active' - Account is active
- 'Inactive' - Account is disabled

6. FARE_RATE Table

Column Name	Data Type	Size	Allow Null	Default	Description
rate_id	INT	-	NO	AUTO_INCREMENT	Unique identifier for fare rate
base_fare	DECIMAL	10,2	NO	-	Base fare amount in PHP
per_km_rate	DECIMAL	10,2	NO	-	Per kilometer rate in PHP
effective_date	DATE	-	NO	-	Date fare rate becomes effective
end_date	DATE	-	YES	NULL	Date fare rate expires
is_active	BOOLEAN	-	YES	TRUE	Whether rate is currently active
updated_by	VARCHAR	50	YES	NULL	LGU staff who updated the rate

7. PLATE_REFERENCE Table

Column Name	Data Type	Size	Allow Null	Default	Description
plate_id	INT	-	NO	AUTO_INCREMENT	Unique identifier for plate reference
plate_number	VARCHAR	20	NO	-	Plate number (unique)
vehicle_type	VARCHAR	30	YES	'Tricycle'	Type of vehicle
status	VARCHAR	20	YES	'Active'	Plate status
first_seen	DATETIME	-	NO	CURRENT_TIMESTAMP	First time plate was reported

Column Name	Data Type	Size	Allow Null	Default	Description
last_seen	DATETIME	-	NO	CURRENT_TIMESTAMP	Last time plate was reported
report_count	INT	-	YES	0	Total number of reports for this plate

Status Values:

- 'Active' - Plate is active
 - 'Inactive' - Plate is inactive
 - 'Flagged' - Plate has been flagged
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8. AUDIT_LOG Table

Column Name	Data Type	Size	Allow Null	Default	Description
log_id	INT	-	NO	AUTO_INCREMENT	Unique identifier for audit log
staff_id	VARCHAR	50	YES	NULL	LGU staff who performed the action
action	VARCHAR	50	NO	-	Type of action performed
details	TEXT	-	YES	NULL	Additional details about the action
timestamp	DATETIME	-	NO	CURRENT_TIMESTAMP	Time action was performed

Action Types:

- 'LOGIN' - Staff logged in
- 'LOGOUT' - Staff logged out
- 'UPDATE_REPORT' - Report status updated
- 'UPDATE_FARE_RATES' - Fare rates updated

- 'ADD_STAFF' - New staff added
 - 'DELETE_STAFF' - Staff deleted
 - 'EXPORT_REPORT' - Reports exported
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Data Flow Dictionary

Data Flow Name	Source	Destination	Description
Trip Data	TRIP	REPORT	Trip details attached to report
Report Status	REPORT	LGU_STAFF	Status displayed to LGU
Fare Rate	FARE_RATE	TRIP	Used for fare computation
GPS Points	GPS_LOG	TRIP	Collected during trip
Plate Number	REPORT	PLATE_REFERENCE	Tracked for history
Audit Log	LGU_STAFF	AUDIT_LOG	System activity tracking
Passenger Info	PASSENGER	TRIP	Passenger's trip history

How to Create the Document

Save as Word Document

1. Open Microsoft Word or Google Docs
2. Create a new document
3. Copy all the Data Dictionary content
4. Format as tables
5. Save as: 07_Data_Dictionary.docx

Save as Markdown (Recommended for GitHub)

In VS Code:

1. Right-click on docs/ folder → New File
 2. Name it: 07_Data_Dictionary.md
 3. Copy and paste the content above
 4. GitHub will render it nicely
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What to Upload

- text
 - diagrams/
 - └─ 06_ERD.png
 - docs/
 - └─ 07_Data_Dictionary.docx
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Summary: All Tables

#	Table	Purpose	Primary Key	Foreign Keys
1	PASSENGER	Passenger accounts	passenger_id	-
2	TRIP	Trip records	trip_id	passenger_id
3	GPS_LOG	GPS route data	log_id	trip_id
4	REPORT	Passenger complaints	report_id	trip_id, updated_by
5	LGU_STAFF	LGU staff accounts	staff_id	-
6	FARE_RATE	Fare rate history	rate_id	updated_by
7	PLATE_REFERENCE	Plate tracking	plate_id	-
8	AUDIT_LOG	System audit	log_id	staff_id